

STUDY QUESTIONS 4 - ANSWERS

1. Corner Point Solution Method-

$$\text{minimize } z = 6x + 5y$$

subject to

$$3x - y \leq 6 \quad (C1)$$

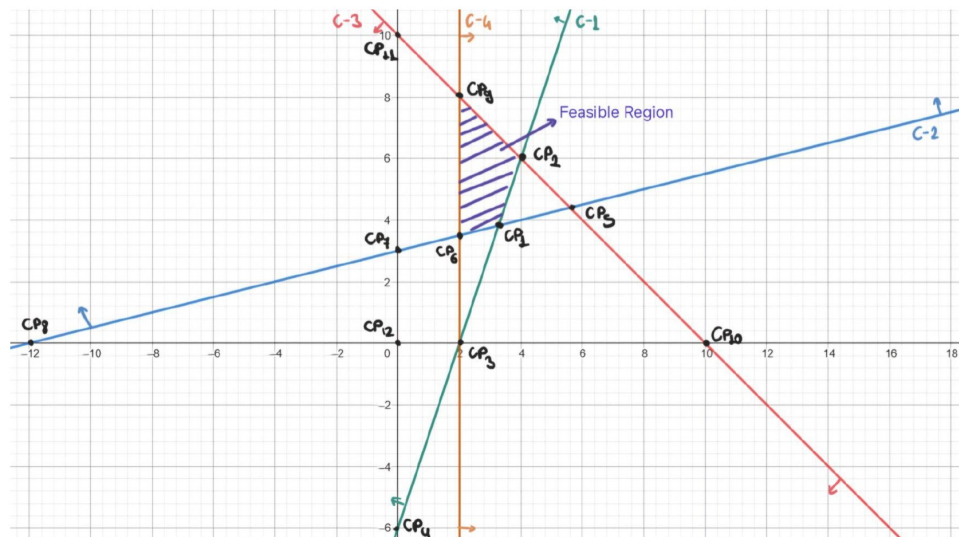
$$-x + 4y \geq 12 \quad (C2)$$

$$x + y \leq 10 \quad (C3)$$

$$x \geq 2 \quad (C4)$$

$$x, y \geq 0 \quad (C5)$$

a)



b)

$$\left(\begin{array}{c} \# \text{ of Constraints} \\ \# \text{ of Decision Variables} \end{array} \right) = \binom{6}{2} = \frac{6!}{4! \cdot 2!} = 15$$

c)

Corner Point	(x, y)	Feasibility	Objective Value (z)
CP-1 (C1 - C2)	(3.27, 3.81)	Feasible	38.67
CP-2 (C1 - C3)	(4, 6)	Feasible	54
CP-3 (C1 - C4 - C5)	(2, 0)	Infeasible	-
CP-4 (C1 - C6)	(0, -6)	Infeasible	-
CP-5 (C2 - C3)	(5.6, 4.4)	Infeasible	-
CP-6 (C2 - C4)	(2, 3.5)	Feasible	29.5
CP-7 (C2 - C6)	(0, 3)	Infeasible	-
CP-8 (C2 - C5)	(-12, 0)	Infeasible	-
CP-9 (C3 - C4)	(2, 8)	Feasible	52
CP-10 (C3 - C5)	(10, 0)	Infeasible	-
CP-11 (C3 - C6)	(0, 10)	Infeasible	-
CP-12 (C5 - C6)	(0, 0)	Infeasible	-

Objective Point $\rightarrow (2, 3.5)$ at Intersection of C2 and C4 $\rightarrow$ Objective Value = 29.5

- d) If we remove C3 feasible region becomes unbounded. However, we are minimizing our mathematical model and we would still obtain 29.5 as our objective value at corner point $(2, 3.5)$. If the problem were maximization problem objective value could get infinitely large without any bound.

2. Graphical Solution Method-

maximize $z = 5x + y$

subject to

$$2x + 3y \leq 12 \quad (C1)$$

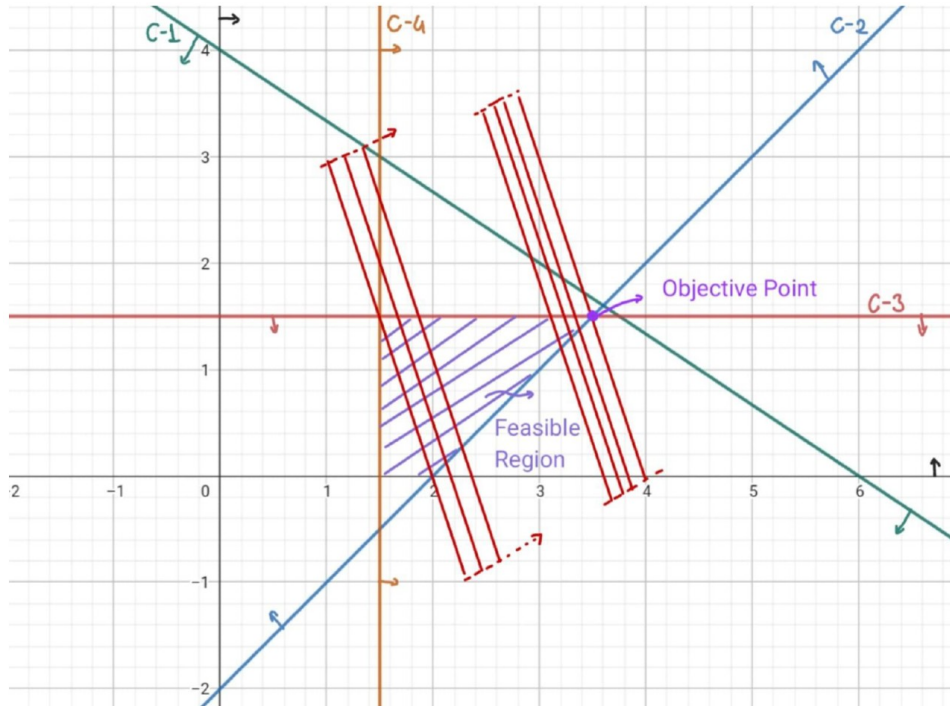
$$x - y \leq 2 \quad (C2)$$

$$2y \leq 3 \quad (C3)$$

$$2x \geq 3 \quad (C4)$$

$$x, y \geq 0 \quad (C5)$$

a)



As it can be seen from the representation, the model is feasible and its feasible region is highlighted. Objective function contour is also highlighted in red. We will slide the contour in the direction that maximizes the objective value. The last point that contour touches before leaving the feasible region is the objective point:

Objective Point $\rightarrow (3.5, 1.5)$ at Intersection of C2 and C3 $\rightarrow$ Objective Value = 19

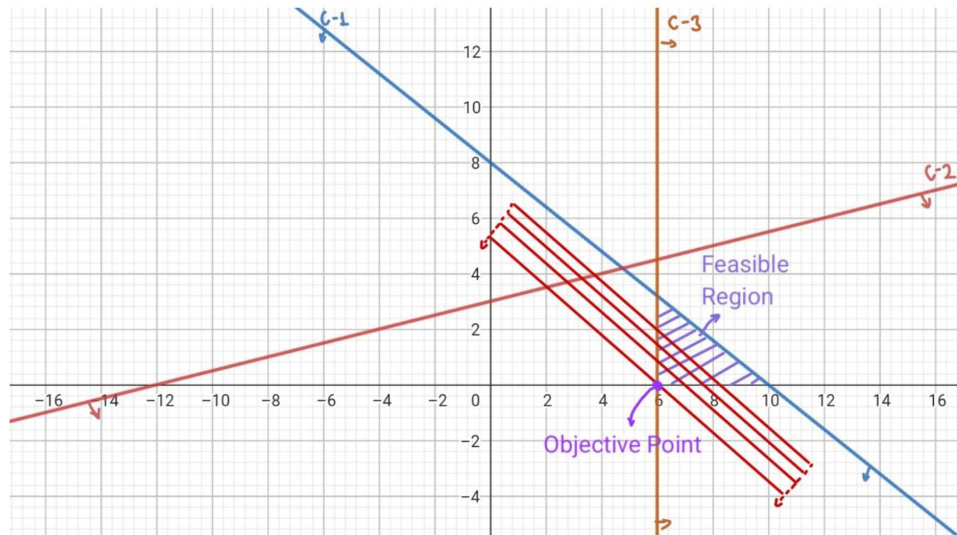
- b) If we multiplied constraint C1 by some scalar $c \in \mathbb{R}$ and added that constraint to our linear programming model, the new constraint would have the exactly same line as the C1. Thus, nature of the model and its objective solution would NOT have changed. However; since we increased the number of constraints in our model, the upper bound for number of corner points will increase as following:

$$\binom{6}{2} = 15 \rightarrow \binom{7}{2} = 21$$

3. An Interesting Model-

$$\begin{aligned}
 &\text{minimize } z = 6x + 7y \\
 &\text{subject to} \\
 &\quad 4x + 5y \leq 40 \quad (C1) \\
 &\quad -x + 4y \leq 12 \quad (C2) \\
 &\quad x \geq 6 \quad (C3) \\
 &\quad x, y \geq 0 \quad (C4)
 \end{aligned}$$

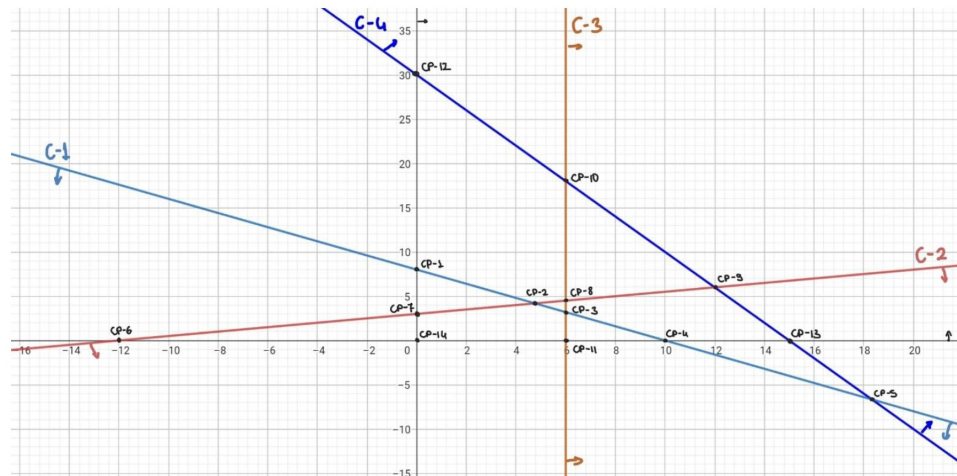
a)



As it can be seen model is feasible and its feasible region is highlighted. Objective function contour is also highlighted in red. We will slide the contour in the direction that minimizes the objective value. The last point that contour touches before leaving the feasible region is the objective point:

Objective Point $\rightarrow (6, 0)$ at Intersection of C3 and C4 $\rightarrow$ Objective Value = 36

b)



With the new constraint the model becomes infeasible; as it can be seen from the graphical representation constraints no longer have intersecting feasible regions. Since we increased the number of constraints in our model, the upper bound for number of corner points will increase as following:

$$\binom{5}{2} = 10 \rightarrow \binom{6}{2} = 15$$

- c) No it is NOT true. A model can have an unbounded feasible region in a direction such that only maximizes objective value, however if that model is a minimization problem then unbounded feasible region wouldn't result in an unbounded objective function. The similar statement can be made for opposite case.

4. A single-server coffee shop system-

	Cust	Arr	Svc Time	Start (max(Arr, Prev.Dep))	Dep (Start+Svc)	Wait (Start-Arr)
a)	1	0	4	0	4	0
	2	3	1	4	5	$4 - 3 = 1$
	3	6	4	6	10	$6 - 6 = 0$
	4	8	2	10	12	$10 - 8 = 2$

Average Waiting Time :

$$\frac{0 + 1 + 0 + 2}{4} = \frac{3}{4} = 0.75 \text{ mins}$$

b) Server Utilization:

- Total Service Time = $4 + 1 + 4 + 2 = 11$ mins.
- Total Time Span (Start $t = 0$ to Last Departure $t = 12$) = 12 mins.
- Utilization $\rho = \frac{11}{12} \approx 91.7\%$.

5. Campus print kiosk system-

Inter-arrival times are cumulative:

$$A_1 = 1, \quad A_2 = 1 + 4 = 5, \quad A_3 = 5 + 1 = 6, \quad A_4 = 6 + 2 = 8, \quad A_5 = 8 + 6 = 14$$

So the arrival times are:

$$A_1 = 1, \quad A_2 = 5, \quad A_3 = 6, \quad A_4 = 8, \quad A_5 = 14$$

The employee can start service only from minute 2.

Service Start Time = $\max(\text{Arrival Time, Previous Departure Time, 2})$

Departure Time = Service Start Time + Service Time

Waiting Time = Service Start Time - Arrival Time

a)

Student	Inter-arrival	Arrival	Service Time	Service Start	Departure	Waiting Time
1	1	1	3	2	5	1
2	4	5	2	5	7	0
3	1	6	5	7	12	1
4	2	8	1	12	13	4
5	6	14	4	14	18	0

b)

Average waiting time:

$$\text{Average Waiting Time} = \frac{1 + 0 + 1 + 4 + 0}{5} = \frac{6}{5} = 1.2 \text{ mins}$$

Maximum number of students waiting in queue:

We track the number of students waiting in queue (arrived but not yet being served) over time:

- Minute 1: Student 1 arrives, waits. Queue=1.
- Minute 2: Student 1 begins service. Queue=0.
- Minute 5: Student 1 departs, Student 2 arrives and immediately begins service. Queue=0.
- Minute 6: Student 3 arrives, Student 2 is still being served (departs at 7). Student 3 waits. Queue=1.
- Minute 7: Student 2 departs, Student 3 begins service. Queue=0.

- Minute 8: Student 4 arrives, Student 3 is still being served (departs at 12). Student 4 waits. Queue=1.
- Minute 12: Student 3 departs, Student 4 begins service. Queue=0.
- Minute 13: Student 4 departs. Queue=0.
- Minute 14: Student 5 arrives and immediately begins service. Queue=0.

The queue never exceeds 1 student at any point in time. Therefore,

$$\boxed{\text{Maximum number of students waiting in queue} = 1}$$

Server utilization:

The observation period runs from minute 0 (kiosk opens) to minute 18 (last student departs):

$$\text{Total elapsed time} = 18 \text{ mins}$$

The server is busy during the following intervals:

- Minutes 2–5: serving Student 1 (3 mins)
- Minutes 5–7: serving Student 2 (2 mins)
- Minutes 7–12: serving Student 3 (5 mins)
- Minutes 12–13: serving Student 4 (1 min)
- Minutes 14–18: serving Student 5 (4 mins)

$$\text{Total busy time} = 3 + 2 + 5 + 1 + 4 = 15 \text{ mins}$$

$$\text{Server utilization} = \frac{\text{Total busy time}}{\text{Total elapsed time}} = \frac{15}{18} \approx 0.833 = 83.3\%$$

6. Advertising and Sales-

a) First, calculate means:

$$\bar{x} = \frac{100 + 200 + 400 + 300}{4} = 250, \quad \bar{y} = \frac{1000 + 1500 + 3000 + 2500}{4} = 2000$$

Calculate S_{xx} and S_{xy} :

$$S_{xx} = \sum (x_i - \bar{x})^2 = (-150)^2 + (-50)^2 + (150)^2 + (50)^2 = 22500 + 2500 + 22500 + 2500 = 50,000$$

$$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$\begin{aligned} S_{xy} &= (-150)(-1000) + (-50)(-500) + (150)(1000) + (50)(500) \\ &= 150,000 + 25,000 + 150,000 + 25,000 = 350,000 \end{aligned}$$

Calculate Slope (β_1) and Intercept (β_0):

$$\beta_1 = \frac{S_{xy}}{S_{xx}} = \frac{350,000}{50,000} = 7$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x} = 2000 - 7(250) = 2000 - 1750 = 250$$

Model: $\hat{y} = 250 + 7x$.

b) Prediction for $x = 150$:

$$\hat{y} = 250 + 7(150) = 250 + 1050 = 1300 \text{ units}$$

7. A real estate company-

The given data are:

House	x_s	x_a	x_c	y
1	1	8	0	180
2	2	12	1	280
3	3	30	0	170
4	4	25	0	210
5	5	20	1	430

where x_s represents size, x_a represents age, x_c represents center, and y represents price.
For each simple linear regression model,

$$\hat{y} = \beta_0 + \beta_1 x$$

we use the following formulas:

$$\beta_1 = \frac{S_{xy}}{S_{xx}}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$R^2 = \frac{S_{xy}^2}{S_{xx}S_{yy}}$$

where

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2$$

calculate the averages:

$$\bar{x}_s = \frac{1+2+3+4+5}{5} = 3$$

$$\bar{x}_a = \frac{8+12+30+25+20}{5} = 19$$

$$\bar{x}_c = \frac{0+1+0+0+1}{5} = 0.4$$

$$\bar{y} = \frac{180+280+170+210+430}{5} = 254$$

Now calculate the deviations from the averages:

House	$y_i - \bar{y}$	$x_{s,i} - \bar{x}_s$	$x_{a,i} - \bar{x}_a$	$x_{c,i} - \bar{x}_c$
1	-74	-2	-11	-0.4
2	26	-1	-7	0.6
3	-84	0	11	-0.4
4	-44	1	6	-0.4
5	176	2	1	0.6

The squared deviations are:

House	$(y_i - \bar{y})^2$	$(x_{s,i} - \bar{x}_s)^2$	$(x_{a,i} - \bar{x}_a)^2$	$(x_{c,i} - \bar{x}_c)^2$
1	5476	4	121	0.16
2	676	1	49	0.36
3	7056	0	121	0.16
4	1936	1	36	0.16
5	30976	4	1	0.36

The products of deviations are:

House	$(x_{s,i} - \bar{x}_s)(y_i - \bar{y})$	$(x_{a,i} - \bar{x}_a)(y_i - \bar{y})$	$(x_{c,i} - \bar{x}_c)(y_i - \bar{y})$
1	148	814	29.6
2	-26	-182	15.6
3	0	-924	33.6
4	-44	-264	17.6
5	352	176	105.6

Therefore, the required sums are:

Term	Value
S_{yy}	46120
$S_{xx,s}$	10
$S_{xx,a}$	328
$S_{xx,c}$	1.2
$S_{xy,s}$	430
$S_{xy,a}$	-380
$S_{xy,c}$	202

- a) • (Size) For the first employee, the model is:

$$\hat{y} = \beta_0 + \beta_1 x_s$$

The slope is:

$$\beta_1 = \frac{S_{xy,s}}{S_{xx,s}} = \frac{430}{10} = 43$$

The intercept is:

$$\beta_0 = \bar{y} - \beta_1 \bar{x}_s = 254 - 43(3) = 125$$

Thus, the estimated regression model is:

$$\boxed{\hat{y} = 125 + 43x_s}$$

Now calculate R^2 :

$$R_s^2 = \frac{S_{xy,s}^2}{S_{xx,s}S_{yy}} = \frac{430^2}{10(46120)} = \frac{184900}{461200} \approx 0.401$$

- (Age) For the second employee, the model is:

$$\hat{y} = \beta_0 + \beta_1 x_a$$

The slope is:

$$\beta_1 = \frac{S_{xy,a}}{S_{xx,a}} = \frac{-380}{328} \approx -1.159$$

The intercept is:

$$\beta_0 = \bar{y} - \beta_1 \bar{x}_a = 254 - (-1.159)(19) \approx 276.01$$

Thus, the estimated regression model is:

$$\boxed{\hat{y} = 276.01 - 1.159x_a}$$

Now calculate R^2 :

$$R_a^2 = \frac{S_{xy,a}^2}{S_{xx,a}S_{yy}} = \frac{(-380)^2}{328(46120)} = \frac{144400}{15127360} \approx 0.010$$

- (Center) For the third employee, the model is:

$$\hat{y} = \beta_0 + \beta_1 x_c$$

The slope is:

$$\beta_1 = \frac{S_{xy,c}}{S_{xx,c}} = \frac{202}{1.2} \approx 168.33$$

The intercept is:

$$\beta_0 = \bar{y} - \beta_1 \bar{x}_c = 254 - 168.33(0.4) \approx 186.67$$

Thus, the estimated regression model is:

$$\hat{y} = 186.67 + 168.33x_c$$

Now calculate R^2 :

$$R_c^2 = \frac{S_{xy,c}^2}{S_{xx,c}S_{yy}} = \frac{202^2}{1.2(46120)} = \frac{40804}{55344} \approx 0.737$$

Model	Estimated Regression Model	R^2
Size Model	$\hat{y} = 125 + 43x_s$	0.401
Age Model	$\hat{y} = 276.01 - 1.159x_a$	0.010
Center Model	$\hat{y} = 186.67 + 168.33x_c$	0.737

- Based on the R^2 values, the center model (employee 3) gives the best fit among the three simple linear regression models.
- I would prefer Model SC. It has a higher R^2 ($0.899 > 0.737$), indicating it explains a larger proportion of the variance in housing prices.
- Yes. Adding variables to a linear regression model can only increase R^2 or keep it constant; it cannot decrease it. Since a model with Size, Age, and Center contains the predictors of both A and B, its R^2 will be ≥ 0.899 .

8. Supplier selection-

- Decision variables:** Let $I = \{A, B, C\}$ be the set of suppliers.

x_i = Number of units purchased from Supplier i

$$y_i = \begin{cases} 1 & \text{if supplier } i \in I \text{ is selected} \\ 0 & \text{otherwise} \end{cases}$$

where:

$$\begin{aligned} x_i &\geq 0 \quad \forall i \in I \\ y_i &\in \{0, 1\} \quad \forall i \in I \end{aligned}$$

- Objective function:

c_i = Unit cost for supplier i

f_i = Fixed cost for supplier i

$$\min Z = \sum_{i \in I} c_i x_i + \sum_{i \in I} f_i y_i \quad \text{OR} \quad 5x_A + 6x_B + 4x_C + 700y_A + 300y_B + 900y_C$$

Demand satisfaction constraint:

$$\sum_{i \in I} x_i = 500 \quad OR \quad x_A + x_B + x_C = 500$$

Capacity and only-buy-if-selected linking constraints: Supplier A can provide at most 300 units. Supplier B can provide at most 250 units. Supplier C can provide at most 400 units. These constraints also ensure that if a supplier is not selected, no purchase can be made from that supplier.

U_i = Capacity for supplier i

$$x_i \leq U_i y_i \quad \forall i \in I$$

OR

$$x_A \leq 300 y_A$$

$$x_B \leq 250 y_B$$

$$x_C \leq 400 y_C$$

Minimum purchase constraints: If a supplier is selected, at least 100 units must be purchased from that supplier.

$$x_i \geq 100 y_i \quad \forall i \in I$$

OR

$$x_A \geq 100 y_A$$

$$x_B \geq 100 y_B$$

$$x_C \geq 100 y_C$$

Variable type constraints: Variable domain restrictions

$$x_i \geq 0 \quad \forall i \in I$$

$$y_i \in \{0, 1\} \quad \forall i \in I$$

9. Distribution center opening problem-

a) Decision variables:

Let $I = \{1, 2, 3\}$ be the set of distribution centers.

Let $J = \{A, B, C, D\}$ be the set of stores.

x_{ij} = amount shipped from DC i to Store j

$$y_i = \begin{cases} 1, & \text{if DC } i \in I \text{ is opened} \\ 0, & \text{otherwise} \end{cases}$$

where:

$$x_{ij} \geq 0 \quad \forall i \in I, \forall j \in J$$

$$y_i \in \{0, 1\} \quad \forall i \in I$$

b)

Objective function:

Let F_i be the fixed opening cost of DC i .

Let c_{ij} be the unit transportation cost from DC i to Store j .

$$\min Z = \sum_{i \in I} F_i y_i + \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij}$$

OR

$$\begin{aligned} \min Z = & 650y_1 + 500y_2 + 300y_3 \\ & + 4x_{1A} + 7x_{1B} + 6x_{1C} + 5x_{1D} \\ & + 6x_{2A} + 4x_{2B} + 5x_{2C} + 8x_{2D} \\ & + 8x_{3A} + 6x_{3B} + 4x_{3C} + 3x_{3D} \end{aligned}$$

Demand satisfaction constraints:

Each store's demand must be satisfied.

$$\sum_{i \in I} x_{ij} = d_j \quad \forall j \in J$$

OR,

$$x_{1A} + x_{2A} + x_{3A} = 30$$

$$x_{1B} + x_{2B} + x_{3B} = 35$$

$$x_{1C} + x_{2C} + x_{3C} = 25$$

$$x_{1D} + x_{2D} + x_{3D} = 25$$

Capacity and linking constraints: These constraints ensure that if a DC is not opened, no shipment can be made from that DC. They also ensure that the capacity of an opened DC is not exceeded.

Let U_i be the capacity of DC i .

$$\sum_{j \in J} x_{ij} \leq U_i y_i \quad \forall i \in I$$

OR,

$$x_{1A} + x_{1B} + x_{1C} + x_{1D} \leq 70y_1$$

$$x_{2A} + x_{2B} + x_{2C} + x_{2D} \leq 60y_2$$

$$x_{3A} + x_{3B} + x_{3C} + x_{3D} \leq 45y_3$$

Variable type constraints:

$$x_{ij} \geq 0 \quad \forall i \in I, \forall j \in J$$

$$y_i \in \{0, 1\} \quad \forall i \in I$$

c) **Feasibility of opening exactly two DCs based only on capacity:**

First, calculate total demand:

$$30 + 35 + 25 + 25 = 115$$

Now check each pair of DCs.

$$\text{DC 1} + \text{DC 2 capacity} = 70 + 60 = 130$$

Since

$$130 \geq 115$$

opening DC 1 and DC 2 is feasible based on capacity.

$$\text{DC 1} + \text{DC 3 capacity} = 70 + 45 = 115$$

Since

$$115 \geq 115$$

opening DC 1 and DC 3 is feasible based on capacity.

$$\text{DC 2} + \text{DC 3 capacity} = 60 + 45 = 105$$

Since

$$105 < 115$$

opening DC 2 and DC 3 is not feasible based on capacity.

Therefore,

Feasible pairs: (1, 2) and (1, 3)

Infeasible pair: (2, 3)

10. **Delivery route planning-**

a) **Nearest neighbor heuristic:**

Start from location A.

Current location: A

From A, the nearest unvisited location is B:

$$d_{AB} = 11$$

Therefore,

$$A \rightarrow B$$

From B, the nearest unvisited location is E:

$$d_{BE} = 9$$

Therefore,

$$A \rightarrow B \rightarrow E$$

From E, the nearest unvisited location is C:

$$d_{EC} = 11$$

Therefore,

$$A \rightarrow B \rightarrow E \rightarrow C$$

From C, the nearest unvisited location is F:

$$d_{CF} = 7$$

Therefore,

$$A \rightarrow B \rightarrow E \rightarrow C \rightarrow F$$

From F, the only remaining unvisited location is D:

$$d_{FD} = 10$$

Therefore,

$$A \rightarrow B \rightarrow E \rightarrow C \rightarrow F \rightarrow D$$

Finally, return from D to A:

$$d_{DA} = 22$$

b) **Initial tour and total distance:**

The initial tour found by the nearest neighbor heuristic is:

$$A \rightarrow B \rightarrow E \rightarrow C \rightarrow F \rightarrow D \rightarrow A$$

Total distance:

$$d_{AB} + d_{BE} + d_{EC} + d_{CF} + d_{FD} + d_{DA}$$

$$= 11 + 9 + 11 + 7 + 10 + 22$$

$$= 70$$

Therefore,

$$\boxed{\text{Initial tour distance} = 70}$$

c) **2-opt move:**

The proposed 2-opt move removes the following two edges:

$$B - E \quad \text{and} \quad D - A$$

Old edge cost:

$$d_{BE} + d_{DA} = 9 + 22 = 31$$

The proposed move adds the following two edges:

$$B - D \quad \text{and} \quad E - A$$

New edge cost:

$$d_{BD} + d_{EA} = 15 + 13 = 28$$

Since

$$28 < 31$$

the 2-opt move gives an improvement.

Improvement amount:

$$31 - 28 = 3$$

d) **Improved tour:**

Since the new edge cost is smaller than the old edge cost, the 2-opt move is accepted.

The initial tour was:

$$A \rightarrow B \rightarrow E \rightarrow C \rightarrow F \rightarrow D \rightarrow A$$

After removing $B - E$ and $D - A$, and adding $B - D$ and $E - A$, the improved tour becomes:

$$A \rightarrow E \rightarrow C \rightarrow F \rightarrow D \rightarrow B \rightarrow A$$

New total distance:

$$70 - 31 + 28 = 67$$

Equivalently,

$$d_{AE} + d_{EC} + d_{CF} + d_{FD} + d_{DB} + d_{BA}$$

$$= 13 + 11 + 7 + 10 + 15 + 11$$

$$= 67$$

Therefore,

Improved tour distance = 67
